Mahmoud Abumandour

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Education

Simon Fraser University — **Burnaby, Canada**
PhD in Computer Science — ***Sep 2024 – Present***
MSc in Computer Science | Thesis: Uncovering and Mitigating Bit-Flip Vulnerabilities in Edge DNNs — ***Sep 2022 – Aug 2024***

Mansoura University — **Mansoura, Egypt**
BSc in Computer and Communication Engineering | Ranked 1st of 180 students — ***Sep 2017 – Jul 2022***

Experience

Simon Fraser University — **Burnaby, Canada**
Graduate Research Assistant — ***Sep 2022 – Present***

- Designed a data-independent bit-flip attack that degrades modern LLMs to 0% accuracy with fewer than 25 bit flips
- Designed a bit-flip protection for LLMs that eliminates white-box bit-flip attacks on GPUs
- Teaching Assistant for Intro to Computer Systems, Compiler Design, Computer Architecture, and Distributed Systems

Max Planck Institute for Security and Privacy — **Bochum, Germany**
Research Intern — ***Jun 2026 – Sep 2026***

- Developed an FPGA-based fault-injection framework to evaluate the security of hardened cores
- Uncovered and reported multiple security vulnerabilities in the OpenTitan ecosystem

Intel Corporation — **Santa Clara, CA (Remote)**
CPU Architecture Intern — ***Jan 2024 – May 2024***

- Modeled and evaluated CPU front-end features, including instruction prefetching and instruction caching
- Characterized workloads by instruction-cache footprint and branch behavior
- Identified sources of miscorrelation between functional and cycle-accurate simulators

Google Summer of Code (RTEMS) — **Remote**
Student Developer — ***May 2022 – Sep 2022***

- Re-architected the release notes generator as a multi-threaded pipeline, achieving an 8x speedup
- Automated release data fetching from the RTEMS bug tracker and Markdown-to-RST/PDF generation

Master Micro — **Cairo, Egypt**
Software Engineering Intern — ***Oct 2021 – Feb 2022***

- Designed a range format for EDA lookup table files, reducing average range-query latency by over 50% vs. a binary format

Google Summer of Code (QEMU) — **Remote**
Student Developer — ***May 2021 – Aug 2021***

- Implemented multi-core, multi-level cache emulation for user-space and full-system workloads

Publications and Talks

- Srinija Ramichetty, **Mahmoud Abumandour**, Alaa R. Alameldeen, Guru Venkataramani. *EnsembleHealer: Autonomous Recovery from Model Poisoning in Decentralized Federated Learning*. ACM Computing Frontiers (CF), 2026.
- **Mahmoud Abumandour**, Srinija Ramichetty, Guru Venkataramani, Alaa R. Alameldeen. *WeightSentry: Real-Time Bit-Flip Protection for Deep Neural Networks on GPUs*. Hardware and Architectural Support for Security and Privacy (HASP), 2025.
- Talk: *Position-Adaptive Temporal Sparsity for KV Caches in Long-Context LLMs*. Sparsity Workshop at MICRO 2025.

Projects and Open-Source Contributions

- **Kyoto Compiler**: Designed the Kyoto language with an LLVM-based compiler and a grammar-based fuzzer in Rust
- **RISC-V Fuzzing Emulator**: Built a RISC-V emulator for fuzzing with linear throughput scaling by avoiding kernel overhead
- **QEMU**: Modernized locking and memory allocation with scope-based APIs. Redesigned the plugin configuration interface
- **SerenityOS**: Defined a global OS versioning API, improved POSIX compliance of utilities, and extended DBMS SQL support
- **RISC-V Newlib**: Profiled and optimized newlib for RISC-V, benchmarking on QEMU, Spike, gem5, and RISC-V hardware
- **FPGA Designs**: Low-power AES core (80% less power than naïve design); hyperthreaded RISC core (5x throughput)

Skills

Languages: C++, C, Rust, Python, x86 Assembly, Bash, Java, SystemVerilog, VHDL
Tools: PyTorch, TensorFlow, LLVM, gem5, QEMU, Ghidra, perf, Git, Docker, Linux, Xilinx Vivado
Areas: Computer Architecture, Hardware Security, ML Security, Fault Injection, Fuzzing, FPGA, RISC-V, ARM